Chapter 5: Materials and Cycles on Earth

Comprehensive Study Notes & Examination Paper

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PART 1: COMPREHENSIVE STUDY NOTES

5.1 The Structure of the Atom

Atoms are the tiny building blocks of all matter. They are made of three **subatomic particles**. The center of the atom is mostly empty space, but contains a dense center called the **nucleus**.

- **Protons:** Positive (+) charge. Packed inside the nucleus. The amount of protons acts as the **atomic number**, which determines exactly what element an atom is. Both protons and neutrons have a mass of 1 atomic unit.
- **Neutrons:** Neutral (0) charge. Also packed inside the nucleus.
- **Electrons:** Negative (-) charge. They have almost zero mass and orbit the nucleus in specific energy levels called **shells**.

Note: Because the number of positive protons exactly equals the number of negative electrons, the entire atom is completely **neutral**.

Historical Models of the Atom

J.J. Thomson's Model ("Plum Pudding"): A solid ball of positive charge with negative electrons stuck inside.
Ernest Rutherford's Model: Discovered through the Gold Foil Experiment. Most alpha particles passed straight through (proving atoms are mostly empty space), but a tiny fraction deflected back (proving the existence of a tiny, incredibly dense, positive nucleus).

5.2 Purity and Alloys

Pure Substances: Made of only one type of atom. They melt at one exact, **sharp** temperature. Pure metals are soft because their neat atomic rows slide easily. Because they are pure elements, they cannot be separated by physical techniques like distillation or **chromatography**.

Alloys: Mixing a metal with other elements creates an **alloy**. Different-sized atoms disrupt the neat rows, preventing them from sliding. This makes them much **harder** and causes them to melt over a wider range of temperatures.

- *Examples:* Stainless steel resists rust, bronze is an ancient favorite, and any alloy containing **Mercury** is uniquely called an amalgam.

5.3 Weather vs. Climate

- **Weather:** The short-term condition of the atmosphere (e.g., today's temperature, **precipitation**, and humidity). It changes rapidly, and extreme anomalies like hurricanes happen within it. Scientists who study daily forecasts are called **meteorologists**.

- **Climate:** The long-term average of weather patterns in an area, scientifically measured over a period of **30 years or more**. Climate is heavily influenced by a region's **latitude** (distance from the equator) and altitude. Massive **ocean currents** also move heat globally.

5.4 Climate and Ice Ages

Earth's climate naturally cycles between **Glacial Periods** (Ice Ages) and Interglacial Periods. During Ice Ages, so much water freezes into glaciers that global **sea levels** drop dramatically, and megafauna like woolly mammoths thrive. These cycles are triggered by natural wobbles in the Earth's **orbit**.

Evidence of Past Climates (Paleoclimatology)

Scientists study past climates using **Ice Cores** (which trap ancient air/CO_2 bubbles) and **Peat Bogs** (acidic soil that preserves ancient plant **pollen**). Additionally, counting **tree rings** reveals how warm and wet specific ancient years were.

5.5 The Atmosphere and Global Warming

- **Ancient Earth:** Volcanoes released high CO_2 and water vapor. There was **no oxygen**.
- **Rise of Oxygen:** Plants evolved and used **photosynthesis**, removing CO_2 and adding O_2. Deforestation removes this natural carbon sink.
- **Modern Era:** Burning fossil fuels violently releases millions of years of locked CO_2. Because CO_2 is a **greenhouse gas**, it traps heat. A natural greenhouse effect shields Earth from freezing, but human activity causes a dangerous *enhanced* effect.

Note: Other powerful greenhouse gases include water vapor and **Methane** (from livestock). Excess CO_2 dissolves into the sea, causing ocean **acidification**. Shifting to **renewable** energies (wind/solar) is the primary solution.

PART 2: O.W.L. EXAMINATIONS

Instructions: Answer all questions in the spaces provided. You may use a calculator.

Section A: Core Concepts

Multiple Choice Questions

Q1. Which subatomic particle has no electrical charge?

A. Proton

B. Electron

C. **Neutron**

D. Nucleus

Q2. What is the dense, central part of an atom called?

A. Shell

B. **Nucleus**

C. Orbit

D. Cloud

Q3. Which of the following is considered a chemically pure substance?

A. 9-carat gold

B. **24-carat gold**

C. Steel

D. Bronze

Q4. Why are impurities often added to pure gold to make jewellery?

A. To make it softer

B. **To make it harder**

C. To make it invisible

D. To decrease its melting point

Q5. Which of the following describes climate rather than weather?

A. A thunderstorm this afternoon

B. A sudden drop in temperature yesterday

C. **The average annual rainfall over 30 years**

D. High humidity levels today

Q6. What do scientists call the coldest periods of an Ice Age where glaciers expand?

A. Interglacial periods

B. Tropical periods

C. **Glacial periods**

D. Greenhouse periods

Q7. Which gas was NOT present in Earth's early atmosphere 4.5 billion years ago?

A. Carbon dioxide

B. Water vapor

C. Nitrogen

D. **Oxygen**

Q8. Which biological process performed by early plants helped reduce carbon dioxide and increase oxygen in the atmosphere?

A. Respiration

B. **Photosynthesis**

C. Combustion

D. Evaporation

Short Answer Questions

Q9. a) Name the subatomic particle that has a negative charge. **Electron**

b) Contrast J.J. Thomson's model with Rutherford's model regarding the center of the atom.

Thomson believed it was a solid positive cloud with scattered electrons. Rutherford discovered a tiny, dense, positive nucleus with mostly empty space.

Q10. Explain the scientific reason why a jeweller suggests buying a 14-carat gold ring instead of a pure 24-carat gold ring for everyday wear.

24-carat gold is pure and soft, so it bends/scratches easily. 14-carat gold is an alloy; the mixed metals disrupt the layers of atoms, making it harder and durable.

Q11. Identify which statement describes Weather and which describes Climate:
A) "It rained heavily this morning, but it is sunny now." **Weather**
B) "The Sahara region is generally hot and dry year-round." **Climate**

Q12. Name the specific plant material preserved in peat bogs that helps scientists figure out what the climate was like thousands of years ago. **Pollen (or pollen grains)**

Q13. State two places where carbon became 'locked up' over millions of years on Earth.

1. **Fossil fuels (coal, oil, gas)** 2. **Rocks (limestone/carbonates)**

Section B: Thinking and Working Scientifically

Multiple Choice Questions

Q14. Why did Rutherford's model of the atom replace Thomson's model?

A. Rutherford was more famous
B. **New experimental evidence showed atoms have a dense nucleus**
C. Thomson's model was too difficult to draw
D. Electrons were discovered to be positive

Q15. A student wants to test if a metal is pure or an alloy. Which physical property would be best to test?

A. Color
B. **Hardness**
C. Smell
D. Size

Q16. A scientist looks at 50 years of continuous temperature data for a city. What is the scientist most likely studying?

A. Today's weather forecast
B. **The local climate trend**
C. The formation of a single hurricane
D. Daily humidity changes

Q17. How do scientists find out about the chemical composition of the atmosphere from thousands of years ago?

A. By reading ancient history books
B. By trapping modern air in jars
C. **By analyzing air bubbles trapped in deep ice cores**
D. By predicting future weather

Q18. What evidence is preserved in peat bogs that tells scientists about past plant life and climates?

A. Dinosaur bones
B. **Trapped pollen grains**
C. Iron ores
D. Liquid water

Q19. A student creates a model of the greenhouse effect using two jars, one covered with a lid and one uncovered, both placed in the sun. What is the independent variable?

A. The temperature of the jars
B. The size of the jars
C. **The presence or absence of the lid**
D. The amount of sunlight

Short Answer Questions

Q20. Why do scientists use physical models to represent atoms? Give one limitation of using a cotton ball and a hula hoop to represent an atom.

Models help visualize things too small to see. Limitation: The scale is highly inaccurate, and atoms are 3D moving structures, not flat or static.

Q21. You have pure iron (soft) and steel (hard alloy). Plan a simple physical test to determine which is which.

Perform a scratch or hardness test. Try to scratch one bar with the other; the alloy (steel) will scratch the pure, softer iron.

Q22. A student measures the temperature rising by 2°C over one week and concludes the climate is permanently getting hotter. Evaluate this conclusion.

The conclusion is invalid. One week is short-term weather. Climate conclusions require average data measured over 30+ years.

Q23. In an ice core, top layers have many CO_2 bubbles; deep/older layers have very few. What conclusion can be drawn from this data?

The amount of Carbon Dioxide in the atmosphere has increased significantly in recent times compared to the distant past.

Q24. As global ice caps melt, less sunlight is reflected and Earth absorbs more heat. Conclude what will happen to the remaining ice caps over the next 50 years.

Temperatures will continue to rise (positive feedback loop), causing the remaining ice caps to melt at an increasingly faster rate.

Section C: Science in Context

Multiple Choice Questions

Q25. Before a new scientific model is published and accepted, it is checked and tested by other scientists. What is this vital process called?

A. **Peer review**
B. Plagiarism
C. Fossilization
D. Climate change

Q26. How has the development of alloys impacted modern human engineering?

A. It has made buildings weaker
B. **It allows for the creation of stronger, durable materials like steel**
C. It has reduced the need for plastics
D. It has stopped the mining of pure metals

Q27. What is the main human activity contributing to the rapid rise in atmospheric carbon dioxide today?

A. Planting trees
B. Building wind turbines
C. **Burning fossil fuels**
D. Recycling plastics

Q28. Which of the following is an example of a renewable energy resource?

A. Coal
B. Natural Gas
C. Oil
D. **Solar power**

Q29. Why is understanding past Ice Ages important for modern society?

A. It helps us breed extinct animals
B. **It provides a baseline to prove how abnormally fast the current climate is changing**
C. It naturally stops global warming
D. It predicts when the next asteroid will hit

Q30. If a global policy banned fossil fuels, what would be the most likely long-term environmental impact?

A. A drastic increase in greenhouse gases
B. **A decrease in the rate of global warming**
C. The destruction of the ozone layer
D. An immediate return to an Ice Age

Short Answer Questions

Q31. Describe how scientists use 'peer review' to verify new ideas before they are accepted by the scientific community.

Independent scientists examine the data, test the experiment themselves, and check for errors to ensure the new model/finding is accurate.

Q32. Give an example comparing the properties of a pure metal to an alloy in construction engineering.

Pure iron is too soft and brittle for large structures. Mixing it with carbon creates steel (an alloy), which is incredibly strong and load-bearing.

Q33. Discuss the global environmental impact of heavily relying on fossil fuels. Provide two consequences.

Consequences include: 1) Enhanced greenhouse effect resulting in global warming. 2) Melting polar ice caps, rising sea levels, and an increase in extreme weather.

Q34. How does studying past Ice Ages help scientists understand current global warming trends?

It establishes a 'baseline' of normal natural temperature cycles. Comparing modern data to this baseline proves the Earth is warming unnaturally fast due to humans.

Q35. If a country switches entirely to renewable energy (wind/solar), explain the effect on the local atmosphere over the next 30 years.

Wind and solar do not release CO_2. Over 30 years, greenhouse gas concentrations would stop rising rapidly, stabilizing the atmosphere and slowing global warming.